Project Title

RAILSAFE ALERT (SMART DESTINATION ALERT SYSTEM)

Project Description

The **RAILSAFE ALERT (SMART DESTINATION ALERT SYSTEM)** is a mobile application that helps passengers avoid missing their destination. The application continuously tracks the train's current location using GPS and compares it with the user's selected destination. It alerts passengers before reaching their station based on either:

- Remaining distance to destination
- Estimated arrival time (ETA)

The application is especially useful for passengers traveling overnight, on long journeys, or on unfamiliar routes.

Team Members

Name	Roll Number
CH. Yaswanth	2024BCS0274
Akash Yadava	2024BCS0250
Prabhas	2024BCS0234
Ankit	2024BCS0262

Problem Statement

Many railway passengers miss their destination because they are sleeping, busy, or unfamiliar with station names. Existing railway applications mainly provide train schedules and live status but do not provide personalized destination alerts.

Our project solves this issue by providing intelligent alerts before the train reaches the selected station.

Objectives

- Prevent passengers from missing their destination.
- Provide customizable alerts.
- Improve passenger convenience.
- Use real-time train tracking.
- Work efficiently with low battery consumption.

Functional Requirements

1. Train Search

- Search train using
    - Train Number
    - Train Name

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2. Destination Selection

- User selects:
    - Boarding station
    - Destination station

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3. Live Train Tracking

- Fetch current train location.
- Display distance remaining.
- Display next station.
- Display ETA.

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4. Destination Alert

User can choose alert type:

- Alert before 10 km
- Alert before 20 km
- Alert before 30 minutes
- Alert before 15 minutes
- Custom alert

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5. Notification System

- Sound notification
- Vibration
- Full-screen popup
- Repeat alert until dismissed

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6. Offline Support

- Store route information.
- Continue alerts even if internet disconnects.

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7. Settings

- Dark Mode
- Language selection
- Alert volume
- Notification preferences

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8. History

- View previous trips.
- Saved favourite routes.

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9. Emergency Feature (Future Enhancement)

- Emergency SOS
- Share live location
- Contact family

Non-Functional Requirements

Performance

- App should respond within **2 seconds**.
- Alert should trigger instantly.
- Real-time updates every few seconds.

Reliability

- 99% alert accuracy.
- Continuous monitoring.
- No missed notifications.

Security

- Encrypted user data.
- Secure authentication.
- Protected user location.

Usability

- Simple UI.
- Easy navigation.
- One-click destination setup.

Scalability

- Support thousands of users simultaneously.
- Cloud database support.

Availability

- 24×7 availability.
- Works on Android devices.

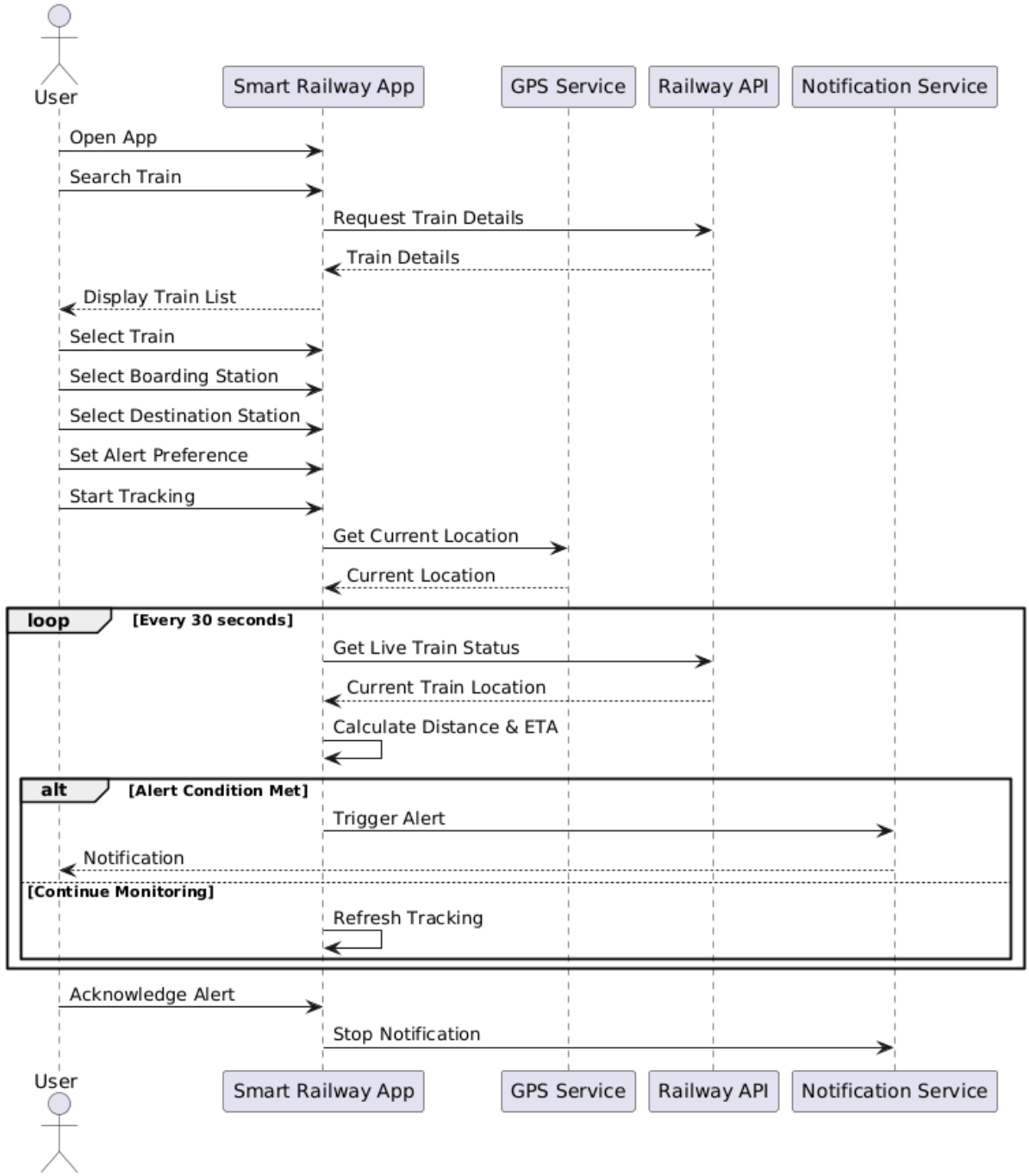
Maintainability

- Modular architecture.
- Easy feature updates.

Portability

- Android support initially.
- Future iOS support.

Sequence Diagram:-



Software Requirements

- Android Studio
- Flutter / React Native
- Firebase Authentication
- Firebase Cloud Messaging
- Google Maps API
- GPS Location Services
- Railway Live Status API
- Cloud Firestore

Hardware Requirements

Minimum Android Phone

- Android 8+
- 3GB RAM
- GPS
- Internet Connection

Development Machine

- Windows 10/11
- 8GB RAM
- VS Code / Android Studio

Existing System

Examples:

- IRCTC Rail Connect
- Where is My Train
- RailYatri
- NTES

Limitations

- Mainly provide train status.
- No personalized destination reminder.
- Limited customization of alerts.
- Alerts may not work offline.

Our Competition

Competitor	Features
IRCTC Rail Connect	Ticket booking, train info
Where is My Train	Offline train status
RailYatri	Live train tracking
NTES	Official train enquiry

How Our Project Differs

Existing Apps	Our Project
General train information	Personalized destination alerts
Manual checking required	Automatic notifications
Limited customization	Distance & time-based alerts
No repeated wake-up alerts	Continuous alerts until dismissed
Basic tracking	Smart prediction using ETA
Mostly online	Partial offline support
No trip reminders	Favourite journeys & history

Modules

1. User Module
2. Train Search Module
3. GPS Tracking Module
4. Alert Engine
5. Notification Module
6. Database Module
7. Settings Module

Role of Each Team Member

Ch. Yashwanth (2024BCS0274)

Team Lead & Backend Development

Responsibilities:

- Project planning
- Requirement analysis
- Backend development
- API integration
- Database design
- Documentation review

Akash Yadav (2024BCS0250)

Frontend Developer

Responsibilities:

- Mobile UI Design
- Train search pages
- Destination selection
- Settings page

Ankit Kumar (2024BCS0262)

GPS & Alert Module

Responsibilities:

- GPS integration
- Distance calculation
- ETA calculation
- Alert generation
- Notification logic

G.SRIRAM PRABHAS(2024BCS0234)

Testing & Documentation

Responsibilities:

- Software testing
- Bug fixing
- Documentation
- UML diagrams
- Presentation preparation

Advantages

- Prevents passengers from missing stations.
- Saves time and reduces travel anxiety.
- Personalized notifications.
- Easy to use.
- Low battery consumption.
- Supports offline route information.
- Useful for elderly passengers and night travel.

Future Scope

- Smartwatch notifications.
- Voice assistant integration.
- AI-based delay prediction.
- Automatic platform information.
- Family trip tracking.
- Multi-language voice alerts.
- Integration with Indian Railways for real-time updates.
- AI-based route recommendations.

Conclusion

The **Smart Railway Alert System** is an intelligent mobile application designed to enhance passenger convenience and safety by providing timely, personalized destination alerts. Unlike existing railway applications that focus primarily on train schedules and live status, our solution proactively notifies users before they reach their destination using real-time GPS tracking, distance calculations, and estimated arrival time. With customizable alerts, offline support, and a user-friendly interface, the system minimizes the risk of passengers missing their station and offers a smarter, more reliable travel experience. This project has the potential to become a valuable companion for daily commuters and long-distance railway travelers.